

MEPCO SCHLENK ENGINEERING COLLEGE (AUTONOMOUS), SIVAKASI
AFFILIATED TO ANNA UNIVERSITY, CHENNAI

Institute Vision	Institute Mission
Envisioning a World Lead by our Engineers, Holding a Beacon of Hope and Confidence for Generations to come	To Produce Competent, Disciplined, Quality Engineers & Administrators Through Service par Excellence

REGULATIONS: MEPCO-R2023 [Common to all B.E. / B.Tech. Degree
(8 – Semester / 6 - Semester LES) Full-Time Programmes] (w.e.f. 2023 – 2024)

CHOICE BASED CREDIT SYSTEM (CBCS)

Degree of Bachelor of Engineering / Bachelor of Technology

1. PRELIMINARY DEFINITIONS AND NOMENCLATURE

In these Regulations, unless the context otherwise requires:

- I. "Programme" means Degree Programme, that is, B.E. / B.Tech. Degree Programme.
- II. "Discipline" means Branch or specialization of B.E. / B.Tech. Degree Programme, like Civil Engineering, Information Technology, etc.
- III. "Course" means a theory or a laboratory subject that is normally studied in a semester, like Mathematics, Physics, Engineering Graphics, and Circuit Theory etc.
- IV. "Chairperson, Academic Council" means the authority of the Autonomous Institution who is responsible for all academic activities of the Institute / Departments for implementation of relevant Rules and Regulations.
- V. "Controller of Examinations" means the authority of the Autonomous Institute who is responsible for all activities of the End semester Examinations.
- VI. "Head of the Institution" means the Principal of the campus.
- VII. "Chairperson, BoS" means Chairperson of Board of Studies of each faculty.
- VIII. "Head of the Department" means Head of the Department concerned.
- IX. "Credit" means a numerical value allocated for each course to describe the student's workload required per week.
- X. "Grade" means the letter grade assigned to each course based on the range of marks specified.
- XI. "Grade Point" means a numerical value (0 to 10) allocated based on the grade assigned to each course.
- XII. "University" means Anna University, Chennai.
- XIII. "LES" means Lateral Entry Scheme.
- XIV. "CBCS" means Choice Based Credit System.
- XV. Honours program normally refers to a higher level of academic achievement in the emerging areas of the same discipline chosen by the students.
- XVI. Minor program normally refers to a higher level of academic achievement in the emerging areas of the other discipline preferred by the students.
- XVII. "Course Instructor" means the faculty of the concerned theory or laboratory course who handles the course.
- XVIII. "Course Coordinator" means the faculty of the concerned theory or laboratory course who coordinates the entire course.
- XIX. "Core Course" means the mandatory courses pertaining to the chosen branch of the students.

- XX. "Core Elective" (CE) course means the courses which are applicable only for the concerned programme students.
- XXI. "Theory Course with Laboratory Component" (TCLC) means a laboratory based theory course.
- XXII. "Open Elective" (OE) course means the courses which are open to all the UG programme students.
- XXIII. "Allied Elective" (AE) means a course which can be opted by students of UG programmes other than students of the hosting programme
- XXIV. "Value Added Course" (VAC) means a course offered by the department to bridge the gap between the curriculum and the requirements of the industry, which could be selected by any student in the college.
- XXV. "Special Course" (SPC) means a course for the registered students of concerned programme, to be offered by experts from reputed industry/institution in India/Abroad.
- XXVI. "Online Course" (OLC) means a course that would be offered by premier institutions like IITs, IISc, etc. through online platform (NPTEL / GIAN / MOOCS / COURSERA, etc.) suggested by the concerned Department.
- XXVII. "Internship" (IS) or Fellowship means a period of time that the student would be deputed to undergo in-house training in reputed Industries or premier institutions or R & D organizations.
- XXVIII. Ability Enhancement Courses (AEC) are the courses for all disciplines which lead to knowledge enhancement in aptitude, coding and soft skills.
- XXIX. Employability Enhancement Courses (EEC) refers to the courses that bridge the gap between skills possessed by the students and the abilities that are required by the organizations.
- XXX. Mandatory Courses (MC) are training programs / courses conducted to enhance the integrity among the students.
- XXXI. "Audit Course" (AUC) means a course which could be audited by the students across any programme in the college.

2. ADMISSION PROCEDURE

2.1.a) Candidates for admission to the first semester of the eight semesters B.E. / B.Tech. Degree Programme shall be required to have passed:

- i. The Higher Secondary Examination of (10+ 2) Curriculum (Regular Academic Stream) prescribed by the Government of Tamil Nadu with Mathematics, Physics and Chemistry as three of the four subjects of the study prescribed under part-III or any other examinations of any University or authority accepted by the Syndicate of the University / Directorate of Technical Education as equivalent thereto.

(OR)

- ii. Should have passed the Higher Secondary Examination of Vocational Stream (Vocational groups in Engineering / Technology) as prescribed by the Government of Tamil Nadu.

2.1.b) Candidates for admission to the second year (LES) of the four year B.E. / B.Tech. Degree Programme shall be required to have passed:

- i. The Diploma Examination with at least 45 % marks (40 % marks in case of candidates belonging to reserved category) in any branch of Engineering / Technology conducted by the State Board of Technical Education and Training, Tamil Nadu.

(OR)

- ii. An Examination of any University or Authority, accepted by the Syndicate of the Anna University as equivalent thereto.

(OR)

- iii. Bachelor of Science (B.Sc.) Degree from a recognized University as defined by UGC, with at least 45 % marks (40 % marks in case of candidates belonging to reserved

category) and passed 10+2 examination with Mathematics as one subject. The students belonging to B.Sc stream shall clear Engineering Graphics and Branch Specific Course from II Semester along with II Year Courses.

In addition, any other conditions as notified by the Government of Tamil Nadu will be followed.

- 2.2. The eligibility criteria such as marks, number of attempts and physical fitness shall be as prescribed by the Syndicate of the University / DoTE from time to time.

3. PROGRAMMES OFFERED

A candidate may be offered admission to any one of the programmes / disciplines of study approved by AICTE, New Delhi and affiliated to Anna University, Chennai.

The list of Programmes / disciplines offered by the Institute with total credits is available in Annexure.

4. STRUCTURE OF THE PROGRAMMES

4.1 Every Programme will have a curriculum with syllabi consisting of theory and laboratory courses such as:

1. Humanities and Social Sciences (HS), including Management
2. Basic Sciences (BS) including Mathematics, Physics, Chemistry, Biology
3. Basic Engineering Sciences (BES), including Materials, Workshop, Drawing, Basics of Electrical / Electronics / Mechanical / Chemical / Computer Engineering, Instrumentation, Bio-Technology
4. Core Courses (CC) – Relevant to the chosen specialization/branch
5. Core Electives (CE) – Six numbers of Core Electives relevant to the chosen specialization/ branch are to be selected and will be offered from V semester onwards.
6. Theory Course with Laboratory Component (TCLC) – A theory course with students doing experiments in laboratories associated with theory part.
7. Open Electives (OE) – Four numbers of Open Elective courses from the list prescribed in the curriculum which will be offered from V semester onwards, with one course in each semester.
8. Allied Electives (AE) – Two numbers of Allied Elective courses from the list prescribed in the curriculum which will be offered in V and VI semesters.
9. Project Work, Mini Project, Seminar, Internship, and Fellowship in reputed organization/Industry.
10. Experiential Project Based Learning - An Employability Enhancement Course (EEC) offered in 6th / 7th semesters of B.E. / B.Tech. CSE / IT / ECE Programmes. Such industry supported EEC course will be offered to other programmes as and when notified by Tamilnadu Government / Anna University, Chennai.
11. Online Course (OLC) – At least four online courses which are conducted by premier institutions like IITs, IISc, etc. through online platform (NPTEL / GIAN / MOOCS / COURSERA, etc.) approved by the concerned Department.
12. Value Added Courses (VAC) which may or may not lead to International / National Certification.
13. Special Course – Course conducted beyond curriculum by industry / Academic experts.
14. Integrated Aptitude Skills (IAS) – Four levels of aptitude training program will be conducted during second and third year of their study for the enhancement of student employability.
15. Integrated Technical Training (ITT) – Training program will be conducted during fourth, fifth and sixth semesters to enhance the coding skill of the students on C, C++, JAVA or PYTHON.
16. Integrated Soft Skills Training (IST) – Two levels of soft skills training program will be conducted to enhance the personality of the students during the second and third year of their study.
17. Mandatory Courses – Training programs or courses like Induction program for first year students, Introduction to Women and Gender Studies, Indian / Tamil culture and heritage courses, Disaster Management etc.

4.2 Course registration

The Institute will ensure the enrolment of all the courses that each student is proposing to undergo in the ensuing semester through the college web portal. The courses that a student enrolls in a

particular semester may include all the courses of the current semester and all the courses that the student has not passed in the previous semesters.

4.3 Procedure for the enrollment for the CE, OE and AE:

The students have to register for the Core, Open and Allied Elective courses through web portal as per the procedure laid down by the institute and approved by the Head of the Institute. The selection of Core, Open and Allied Elective courses shall be done in accordance with the prerequisite of the concerned course. The respective Heads of the Departments offering the Elective course will approve the list of students opting the course.

4.4 New courses may be introduced by the Department / Institution at any time depending upon the requirement after getting approval from the concerned Board of Studies, Academic Council and Governing Body.

4.5 Each course is normally assigned a certain number of credits as follows:

- i. 1 credit per lecture period per week and 1 credit per 2 tutorial periods per week for theory courses.
- ii. 1 credit for 2 periods of laboratory courses and Experiential Project Based Learning course per week.
- iii. 1 credit for 2 periods of technical seminar or 2 periods of mini project work per week.
- iv. 1 credit for acquiring an International Certification in a relevant discipline for showing their competency in the international level as suggested and approved by the HOD and BoS, subject to a maximum of 2 credits during their programme.

4.6 Curriculum Structure of a Semester:

The curriculum structure of a semester for any programme shall be designed in any one of the following combinations with a blend of theory courses not exceeding 6 and laboratory courses not exceeding 4, subject to a maximum of 10 courses. In addition, one to four EEC and AEC courses as discussed in Clause 4.9 and 4.10 will be offered during 2 – 6 semesters.

Semester	Theory Courses	TCLC	Mini Project	Laboratory Course	Project Work
1	6	-	-	3	-
2	5	1	-	3	-
3 – 6	6	-	1	2	-
	5	-	1	3	-
	5	1	1	2	-
7	5	-	-	1	1 (Phase 1)
8	2	-	-	-	1 (Phase 2)

4.7 For the award of the degree, a student has to earn at least the minimum number of credits specified in the curriculum of the relevant discipline / branch of study as mentioned in the Annexure.

4.8 The medium of instruction is English for all courses, examinations, seminar presentations and project report except Tamil Language Mandatory Courses.

4.9 Employability Enhancement Courses (EEC):

4.9.1 Communication Skills:

In order to improve the communication skills of students, a practical course on “Communication Skills” is introduced during their II semester with one credit. Students will be evaluated through the end semester external examination for 100 marks.

4.9.2 Presentation and language skills:

In order to enhance the presentation skills of students, which are greatly expected in employments, a practical course on “Presentation and Language Skills Laboratory” is

introduced during III / IV semester with one credit. Students will be evaluated through end semester external examination for 100 marks.

4.9.3 Career planning and professional essentials:

In order to enrich the employability skills of students which are essential for them while attending placement interviews, a practical course on “Career Planning and Professional Essentials Laboratory” is introduced during V / VI semester with one credit. In addition to the employability skills, students are taught with research paper writing and patent drafting procedures. Students will be evaluated through an end semester external examination for 100 marks.

4.10 Ability Enhancement Courses (AEC):

4.10.1 Integrated Aptitude Skills (IAS)

In order to enhance the employability of students, they are given aptitude training on significant areas like Verbal ability, Logical Reasoning and Quantitative aptitude. The courses on Integrated Aptitude Skills (IAS) are assigned with one credit per course.

The credits earned through these Courses will not be considered for CGPA calculation. The performance of the students in these Courses is evaluated through end semester internal examination for 100 marks. If a student gets a pass mark in this Course, then the credits earned will be mentioned in the grade sheet or else, it will not be reflected in the grade sheet.

4.10.2 Integrated Technical Training (ITT)

In order to promote the employability skills of students, they are given technical hands-on training on significant programming environments like Python, C, C++ or JAVA. The choice of the programming language would be updated now and then, based upon the new update or release. The courses on Integrated Technical Training (ITT) are assigned with one and a half credit per course.

The credits earned through these Courses will not be considered for CGPA calculation. The performance of the students in these Courses is evaluated through end semester internal examination for 100 marks. If a student gets a pass mark in this Course, then the credits earned will be mentioned in the grade sheet or else, it will not be reflected in the grade sheet.

4.10.3 Integrated Soft Skills Training (IST)

Integrated Soft Skills Training is intended to equip future Engineering professionals with skills that are often overlooked in conventional education. It aims to bridge the soft skills gap prevalent among the student community. The courses on Integrated Soft Skills Training (IST) are assigned with one credit per course.

The credits earned through these Courses will not be considered for CGPA calculation. The performances of the students in these Courses are evaluated through end semester internal examination for 100 marks. If a student gets a pass mark in this Course, then the credits earned will be mentioned in the grade sheet or else, it will not be reflected in the grade sheet.

5. HONOURS / MINOR PROGRAMS

The Departments will be offering the following Honours / Minor program in emerging areas under the respective UG Degree Programme as given in the table. There will be 6 or more courses under each of the 27 domains. Students are expected to successfully complete any 6 courses additionally and earn 18 credits.

Sl.No	Emerging Areas	Programmes eligible for Honours	Programmes eligible for Minor
1.	Smart Cities	B.E CIVIL	B.E EEE, ECE, CSE, MECH, BME B.Tech IT, BT, AIDS
2.	Infrastructure Engineering		
3.	Construction Technology		
4.	Smart Grid Technologies	B.E EEE	B.E CIVIL, ECE, CSE, MECH, BME B.Tech IT, BT, AIDS
5.	Electrical & Computer Engineering		

Sl.No	Emerging Areas	Programmes eligible for Honours	Programmes eligible for Minor
6.	Control Systems and Sensors Technology	B.E EEE	B.E CIVIL, ECE, CSE, MECH, BME B.Tech IT, BT, AIDS
7.	Internet of Things	B.E EEE, ECE, CSE, BME B.Tech IT, AIDS	B.E CIVIL, MECH B.Tech BT
8.	Sensor Technology	B.E EEE, ECE, CSE, BME B.Tech IT	B.E CIVIL, MECH B.Tech BT, AIDS
9.	VLSI Design & Technology	B.E EEE, ECE, BME	B.E CIVIL, CSE, MECH B.Tech IT, BT, AIDS
10.	5G and Advanced Technologies	B.E ECE	B.E CIVIL, EEE, CSE, MECH, BME B.Tech IT, BT, AIDS
11.	Artificial Intelligence and Machine Learning	B.E ECE, CSE, BME B.Tech IT	B.E CIVIL, EEE, MECH B.Tech BT
12.	Cyber Security	B.E ECE, CSE, BME B.Tech IT, AIDS	B.E CIVIL, EEE, MECH B.Tech BT
13.	3D Printing	B.E CIVIL, MECH	B.E EEE, ECE, CSE, BME B.Tech IT, BT, AIDS
14.	Robotics	B.E EEE, ECE, MECH, BME	B.E CIVIL, CSE B.Tech IT, BT, AIDS
15.	Electric Vehicles	B.E EEE, MECH	B.E CIVIL, ECE, CSE, BME B.Tech IT, BT, AIDS
16.	Energy Engineering	B.E MECH	B.E CIVIL, EEE, ECE, CSE, BME B.Tech IT, BT, AIDS
17.	Systems Engineering		
18.	Data Science	B.E ECE, CSE, BME B.Tech IT	B.E CIVIL, EEE, MECH B.Tech BT
19.	Blockchain	B.E ECE, CSE, BME B.Tech IT, AIDS	B.E CIVIL, EEE, MECH, B.Tech BT
20.	Virtual Reality & Augmented Reality	B.E CSE B.Tech IT, AIDS	B.E CIVIL, EEE, ECE, MECH, BME B.Tech BT
21.	Genome Engineering & Technology	B.Tech BT	B.E CIVIL, EEE, ECE, CSE, MECH, BME B.Tech IT, AIDS
22.	Biosimilar Technology		
23.	Computational Biology		
24.	Medical Robotics and Devices	B.E BME	B.E CIVIL, EEE, ECE, CSE, MECH B.Tech IT, BT, AIDS
25.	Advance Web development	B.E CSE, B.Tech AIDS, IT	B.E EEE, ECE, MECH, BME, B.Tech BT
26.	Innovation, Entrepreneurship and Venture Development	-	B.E CIVIL, EEE, ECE, CSE, MECH, BME B.Tech IT, BT, AIDS
27.	Business Development, Marketing and Finance	-	B.E CIVIL, EEE, ECE, CSE, MECH, BME B.Tech IT, BT, AIDS

5.1 B.E / B.Tech Honours (Specialization in the same Discipline)

Students can undergo any 6 courses as per clause 5.7 belonging to a particular emerging area eligible as Honours for their programme as listed in the above table. After the successful completion of the programme as per clause 17.2.1 or 17.2.2, he / she will get B.E / B.Tech Honours degree with specialization.

For example, students of Civil / Mechanical Engineering, who have studied six additional courses under “3D Printing” will be given degree B.E (Honours) in Civil / Mechanical Engineering with Specialization in 3D Printing.

5.2 B.E / B.Tech Honours

Students who has successfully completed any six additional courses as per Clause 5.7 across the various emerging areas eligible as Honours for their Programme as per clause 17.2.1 or 17.2.3 will get B.E / B.Tech Honours degree. The specialization will not be mentioned in their Degree.

5.3 B.E / B.Tech (Minor in other Specialization)

Students can undergo any 6 courses as per clause 5.7 belonging to a particular emerging area eligible as Minor for their programme as listed in the above table. After the successful completion of the programme as per clause 17.2.1 or 17.2.4, he / she will get B.E / B.Tech degree in the respective branch with Minor Degree in chosen emerging area.

For an example, students of Civil / Mechanical Engineering, who have studied six additional courses from “Internet of Things” will be given degree B.E in Civil / Mechanical Engineering with minor degree in Internet of Things.

5.4 Registration:

The Institute will ensure the display of all the emerging areas with the concerned courses, which any eligible student can opt to undergo in the ensuing semester, through the college web portal.

5.5 Eligibility:

Students with 7.5 CGPA and above with no history of arrears are eligible to register for Honours with specialization and Honours programs during the course of their third or fourth semester. However, interested students with less than 7.5 CGPA can also register with the consent of the HoD.

Students with 6.5 CGPA and above are eligible to register for Minor program during the course of their third or fourth semester.

5.6 Procedure for the Enrollment:

The students have to register for this Honours / Minor program through web portal as per the procedure laid down by the institute and approved by the Head of the Institute. The selection of Honours / Minor program shall be done based upon the interest of the students. The respective Heads of the Departments offering the Honours / Minor program will approve the list of students.

5.7 Curriculum Structure:

The curriculum structure of Honours / Minor program for any UG programme shall be designed to fit the following distribution of 6 theory courses of 3 credits each in addition to the regular courses. The six courses will be offered during the IV to VII semester of the UG programme.

Semester	IV	V	VI	VII
No. of Courses	2	2	1	1

5.8 SWAYAM NPTEL courses for earning credits:

The students can earn a maximum of 6 credits out of 18 credits through online mode preferably through SWAYAM NPTEL platform. The equivalent NPTEL courses and their corresponding credits will be approved by the Department with the prior approval from the Head of the Institution and the Board of Studies and the same can be used to drop only the corresponding Honours / minor course. The online course (NPTEL) should be completed with not less than B+ Grade in advance to drop a corresponding honours course.

5.9 Cancellation of the programme:

- If a student fails in course(s) of regular degree program and successfully completes the course(s) of Honours / Minor in a semester, his/her registration to the Honours / Minor will be cancelled automatically. However, the grades obtained in the successfully completed Honours / Minor course(s) during current and previous semester(s) will be considered in the respective SGPA, eventually considered in CGPA also.
- If a student fails in course(s) of the Honours / Minor and successfully completes the course(s) of his regular degree program in a semester, his/her registration to the Honours / Minor will be cancelled automatically. However, the grades obtained in the successfully completed Honours / Minor course(s) during previous and current semester(s) will be considered in the respective SGPA, eventually considered in CGPA also.

- iii. If a student fails both in course(s) of his regular degree program and course(s) of the Honours / Minor in a semester, his/her registration to the Honours / Minor will be cancelled automatically. However, the grades obtained in the successfully completed Honours / Minor course(s) during previous semester(s) will be considered in the respective SGPA, eventually considered in CGPA also.
- iv. If a student successfully completes both course(s) of his regular degree program and course(s) of the Honours / Minor in a semester, his/her registration to the Honours / Minor will be continued automatically. The grades obtained in the successfully completed Honours / Minor course(s) during current and previous semester(s) will be considered in the respective SGPA, eventually considered in CGPA also. The student will be awarded Honours / Minor as per clause 17.2.
- v. If a student wishes to withdraw from Honours / Minor during any semester, he / she can be permitted but the grades of successfully completed Honours / Minor course(s) during previous semester(s) will be included in the respective SGPA and eventually considered in overall CGPA also.

5.10 Dropping of electives:

The registration of Honours / Minor program shall get cancelled vide clause 5.9 (i – iii) and the student can also withdraw from Honours / Minor program as per clause 5.9 (v). After the cancellation / withdrawal, if the student wishes, he / she can use the credits of the successfully completed Honours / Minor courses to drop any two Allied / Open elective courses as per the following conditions. However, if a student has successfully completed more than two honours / Minor courses at the time of honours / Minor cancellation / withdrawal, such courses will be mentioned in the Semester Grade statement but only the courses with higher Grades will be considered for the final CGPA calculation in the Consolidated Grade statement.

- i. The students who have opted Minor degree program (clause 5.3) can drop only the Allied Elective courses.
 - ii. The students who have opted Honours with specialization / Honours degree program (clause 5.1 / clause 5.2) can drop the Open / Allied Elective courses.
 - iii. Whereas the dropping of regular courses as per clause 29 is not applicable to this program.
- 5.11** New Honours / Minor in emerging areas may be introduced by the department / institution at any time depending upon the requirement after getting approval from the concerned Board of Studies, Academic Council and Governing Body.

6. DURATION OF THE PROGRAMME

- 6.1 A student is ordinarily expected to complete the B.E. / B.Tech. Programme in 8 semesters (four academic years) but in any case, not more than 7 years (14 semesters). In case of LES, a student is ordinarily expected to complete the B.E. / B.Tech. Programme in 6 semesters (three academic years) but in any case, not more than 6 years (12 semesters).
- 6.2 Each semester shall normally consist of 90 working days or 540 periods of 50 or 45 minutes each. The Head of the Department shall ensure that every teacher teaches the entire content of the specified syllabus as per the number of periods specified in the syllabus.
- 6.3 However, special theory/laboratory classes may be conducted for students who require additional coaching over and above the number of periods normally specified, as decided by the class committee.

But for the purpose of calculation of attendance requirements for writing the End Semester Examinations (as per Clause 7) by the students, only 540 periods per semester conducted within the specified academic schedule shall be taken into account.

- 6.4 The total period for completion of the programme reckoned from the commencement of the first semester (third semester in case of LES) to which the candidate was admitted shall not exceed the maximum period specified in Clause 6.1 irrespective of the period of break of study (vide Clause 19.1) or period of prevention in order that he/she may be eligible for the award of the degree (vide Clause 17).

7. REQUIREMENTS FOR COMPLETION OF A SEMESTER

- 7.1 A candidate who has fulfilled the following conditions shall be deemed to have satisfied the requirements for completion of a semester.

Ideally every student is expected to attend all classes and earn 100% attendance. However, in order to allow provision for certain unavoidable reasons such as medical / personal grounds / participation in external events, the students are expected to earn a minimum of 75% attendance. Therefore, he/she shall secure not less than 75% of overall attendance in that semester taking into account the total number of periods in all courses attended by the candidate as against the total number of periods in all courses offered during that particular semester.

- 7.2 However, a candidate who secures attendance between 65% and 75% in the current semester due to medical reasons (hospitalization / accident / specific illness) or due to participation in the University / State / National / International level events with prior permission from the Head of the Department concerned can get exemption from the prescribed attendance requirement and he/she shall be permitted to appear for that particular semester examinations.
- 7.3 Candidates who secure less than 65% overall attendance will not be permitted to write the End-Semester Examinations and are not permitted to go to next / subsequent semester. They have to repeat the incomplete semester in the next academic year.

8. FACULTY ADVISOR

To help the students in planning their courses of study and for general advice on the academic programme, the Head of the Department of the student will attach a certain number of students to a faculty member of the Department who shall function as Faculty Advisor for those students throughout their period of study. Such Faculty Advisor shall advise the students and monitor the courses taken by the students, check the attendance and progress of the students attached to him / her and counsel them periodically. If necessary, the Faculty Advisor may also discuss with or inform the parents about the progress / performance of the students concerned.

The responsibilities of the Faculty Advisor shall be:

- i. To act as the channels of communication among HoD, students of the respective class and parents.
- ii. To collect and maintain various statistical details of academic and other activities of the students.
- iii. To help the chairperson of the class committee in planning and conducting the class committee meetings.
- iv. To monitor the students' academic performance and attendance and to inform the parents.
- v. To encourage the students to win awards, medals, scholarships etc.
- vi. To motivate students to undergo industrial visits, in-plant trainings, internships, fellowships, etc.
- vii. To address the students' grievances.

9. CLASS COMMITTEE

A class committee consists of course handling faculty members of the class concerned, student representatives and a chairperson, who does not handle any subject for the class. If required, a senior faculty member from an allied department can also be the chairperson. It is like the 'Quality Circle' (more commonly used in industries), with the overall goal of improving the teaching-learning process. The functions of the class committee include:

- i. Solving the problems experienced by students in the classroom and in the laboratories in consultation with Head of the Department.
- ii. Clarifying the Regulations of the degree programme and the details of rules therein.
- iii. Informing the student representatives, the academic schedule including the mode and dates of assessments and the syllabus coverage for each assessment.
- iv. Informing the student representatives, the details of Regulations regarding weightage used for each assessment.

- v. Analyzing the performance of the students of the class after each periodic test and finding means and ways of improving the performance of the students.
- vi. Identifying the slow learners and helping them to come up, by guidance and/or by providing additional coaching.

The class committee is normally constituted by the Head of the Department. However, if the students of different branches are mixed in a class, the class committee is to be constituted by the concerned HODs. The class committee shall be constituted within a week from the date of commencement of a semester. At least 6 student representatives from the respective class (typically 3 boys and 3 girls) shall be included in the class committee. The Chairperson of the class committee can invite the Faculty Advisor(s) and the Head of the Department to the meeting of the class committee. The chairperson of the class committee is required to prepare the minutes of every meeting, submit the same to the Head of the Department and then to the Principal within three working days of the meeting and arrange to circulate the same among concerned students and faculty.

The first meeting of the class committee shall be held within fifteen days from the date of commencement of the semester. The nature and weightage of continuous assessments shall be informed in the first meeting, and the same shall be communicated to the students. Two or three subsequent meetings in a semester may be held at suitable intervals. During these meetings the student members representing the entire class, shall meaningfully interact and express their opinions and suggestions of the class students to improve the effectiveness of the teaching-learning process.

10. COURSE COMMITTEE FOR COMMON COURSES

Each common theory course offered to more than one discipline or more than one class handled by different faculty members, shall have a "Common Course Committee" comprising all the teachers teaching the course with one of them nominated as Course Coordinator. The nomination of the Course Coordinator shall be made by the Faculty Chairman in consultation with Heads of Departments of the participating discipline. The "Common Course Committee" shall meet as often as possible and ensure uniform delivery and evaluation of continuous assessments after arriving at a common scheme of evaluation (subject to clause 11). Wherever feasible, the common course committee shall prepare a common question paper for the test(s).

11. PROCEDURES FOR AWARDED MARKS FOR CONTINUOUS ASSESSMENT (CA)

For all B.E. / B.Tech. Programmes, out of 100 marks, the maximum marks for continuous Assessment is fixed as 40 for Theory Courses as well as Theory Courses with laboratory components and End Semester Examination carries 60 marks.

However, continuous assessment is 60 marks for Laboratory Courses and End Semester Examination carries 40 marks. The continuous assessment is fixed as 60 marks for mini project, Project Work (Phase I) while viva-voce examination carries 40 marks. Whereas continuous assessment is 120 marks for the Project Work (Phase II) and viva-voce examination carries 80 marks.

Every faculty member is required to maintain an 'Attendance and Assessment Record' for every semester which consists of attendance marked in each lecture or laboratory or project work class, the test marks, Comprehensive Technical Test marks, assignment marks and the record of class work (topics covered), separately for each course handled by the teacher. This should be submitted to the Head of the Department periodically (at least three times in a semester) for checking the syllabus coverage and the records of test marks, Comprehensive Technical Test marks and assignment marks. The Head of the Department will affix his / her signature and date after due verification. At the end of the semester, the record should be verified by the Head of the Department / Institution who will keep this document in safe custody (for five years).

11.1 Continuous Assessment for Theory Courses

(a) Continuous Assessment tests (75% credence)

Three Assessment tests, each carrying 50 marks shall be conducted by the Department / Institution. The total marks obtained in all the tests put together out of 150, shall be reduced to 75 marks and rounded to the nearest integer. This implies equal weight to all the three tests. However, an improvement test at the discretion of the concerned course

faculty / Head of the Department may be conducted for deserving candidates at the end of the semester, where the students have to register/request for the improvement test.

(b) Comprehensive Technical Test (15% credence)

In order to test the Comprehensive knowledge of the students on the theory course studied in a semester, a Comprehensive Technical Test (CTT) shall be conducted by the course handling faculty member after the end semester examination on the same day. The question paper for the CTT will be in Multiple Choice Questions (MCQ) format requiring higher level of thinking and understanding for answering. The CTT will be conducted in online mode with negative marking.

(c) Assignment (10% credence)

Two assignments, with equal weightage, either written or tutorials or quizzes or case studies or technical presentations shall be given for the enhancement of higher cognitive levels of the students.

11.2 Continuous Assessment for Laboratory Courses

The maximum marks for Continuous Assessment shall be 60 in case of laboratory courses. Every laboratory exercise / experiment shall be evaluated based on conduct of experiment / exercise and laboratory manual maintained. There shall be at least two model tests. The criteria for arriving at the Continuous Assessment marks of 60 are as follows:

Experiment / Laboratory Manual and Observation cum Record / Laboratory Classes performance: 50% credence

Model Tests: 50% credence

11.3 Continuous Assessment for Theory Courses with Laboratory Component (TCLC)

If there is a theory course with laboratory component, there shall be two assessments: the first assessment (maximum mark is 100) will be similar to assessment of theory course and the second assessment (maximum mark is 100) will be similar to assessment of laboratory course.

Theory component (The weighted average shall be converted to 40 marks)

Continuous Assessment tests: 75% credence

Comprehensive Technical Test: 15% credence

Assignment: 10% credence

Laboratory component (The weighted average shall be converted to 60 marks)

Experiment / Laboratory Manual and Observation cum Record / Laboratory Classes performance: 50% credence

Model Tests: 50% credence

11.4 Continuous Assessment for Mini Project

The maximum marks for Continuous Assessment shall be 60 in case of mini project. The criteria for arriving at the Continuous Assessment are as follows:

Continuous Assessment			
Guide	Review I	Review II	Review III
15	10	15	20

11.4.1 The candidate is expected to submit the project report on or before the last working day of the semester.

11.5 Continuous Assessment for Project Work (Phase I)

The maximum marks for continuous Assessment shall be 60 in case of Project Work (Phase I). There shall be three assessments (each 100 marks) during the semester by a review committee. The students of the project group (of not exceeding 3 students) shall make presentation on the

progress made before the committee. The Head of the Department shall constitute the review committee for each domain of study. There shall be a minimum of three members in the review committee. The guide will be an additional member of the Review Committee and he / she should be present during the presentation of his/her group.

The criteria for arriving at the Continuous Assessment are as follows:

Continuous Assessment			
Guide	Review I	Review II	Review III
15	10	15	20

11.5.1 The candidate is expected to submit the project report on or before the last working day of the semester. Also the candidate is expected to present a paper based on the outcome of the project in IEEE / Elsevier / Springer conferences or submit a manuscript to SCI / Scopus Journals.

11.6 Continuous Assessment for Project Work (Phase II)

The maximum marks for continuous Assessment shall be 120 in case of Project Work (Phase II). There shall be three assessments (each 100 marks) during the semester by a review committee. The students of the project group (of not exceeding 3 students) shall make presentation on the progress made before the committee. The Head of the Department shall constitute the review committee for each domain of study. There shall be a minimum of three members in the review committee. The guide will be an additional member of the Review Committee and he / she should be present during the presentation of his/her group.

The criteria for arriving at the Continuous Assessment are as follows:

Continuous Assessment			
Guide	Review I	Review II	Review III
30	20	30	40

11.6.1 The candidate is expected to submit the project report on or before the last working day of the semester. Also the candidate is expected to submit a manuscript based on the outcome of the project to SCI / Scopus Journals.

11.7 Continuous Assessment for Experiential Project Based Learning

The maximum marks for continuous Assessment shall be 100. Essentially, it involves 15 weeks for learning and implementing projects, and one week for evaluation. The evaluation of Experiential Project Based Learning is as follows:

Project Review & Scoring: Evaluator assesses the project deliverables, reviews the work done by the team and assigns the score for defined metrics.

Project Status Review: Mentor reviews the deliverables submitted by student teams and shares his/her comments. Mentor ensures the timely completion of project.

12. EXAMINATIONS

Performance in each course of study shall be evaluated based on (i) Continuous assessment throughout the semester and (ii) End semester examination. Each course, both theory and laboratory shall be evaluated for a maximum of 100 marks. The end semester examination (laboratory and theory) of 3 hours duration shall be conducted between October - December during the odd semesters and between April - June during the even semesters.

12.1 Theory Courses Examination

Each theory course shall be evaluated for a maximum of 100 marks.

Continuous Assessment : 40 Marks

End-Semester Examinations : 60 Marks

12.2 Laboratory courses Examination

Each laboratory course shall be evaluated for a maximum of 100 marks. The performance of the students in the end semester examinations for laboratory courses are assessed by two Examiners viz. Examiner-I, Examiner-II wherein the Examiner-I would be the laboratory course in-charge and Examiner-II would be the faculty member from the panel of expert nominated by the concerned department.

Continuous Assessment : 60 Marks

End-Semester Examinations : 40 Marks

12.3 Theory Courses with Laboratory Component (TCLC) Examination

Each Theory Course with Laboratory Component (TCLC) shall be evaluated separately for theory component and laboratory component. The theory component will be evaluated vide clause 12.1 and the laboratory component vide clause 12.2.

The criteria for arriving at the aggregate end semester examination marks are as follows:

End semester theory examinations : 75% credence

End semester laboratory examinations : 25% credence

12.4 Mini Project

The award of end semester marks for 40 shall be evaluated by both Examiner I and Examiner II, out of which the evaluation of the final report submitted by the student group shall carry a maximum of 10 marks (same marks must be awarded to every student of the project group) while the viva-voce examination shall carry 30 marks (awarded to each student of the mini project group based on the individual performance in the viva-voce examination) as follows:

End Semester Examinations				
Report Evaluation		Viva-Voce		
Examiner I	Examiner II	Examiner I	Examiner II	*Guide
5	5	10	10	10

*In the absence of guide due to unavoidable reasons during the end semester examination, the Examiner 1 will award the marks for the concerned student.

12.4.1 If a student, has not submitted the project work report on or before the last working day of the semester or has failed in the end semester examination for Mini Project Work, then he/she has to re-register for the same course for re-doing the Mini Project Work and appear for the end semester examination in the subsequent semester(s).

12.5 Project Work Examination (Phase I)

The project work (Phase I) shall be evaluated for a maximum of 100 marks each. The Continuous Assessment will be done for 60 marks as per Clause 11.5.

The award of end semester marks for 40 shall be evaluated by both Internal Examiners and External Examiners, out of which the evaluation of the final report submitted by the student group shall carry a maximum of 10 marks (same marks must be awarded to every student of the project group) while the viva-voce examination shall carry 30 marks (awarded to each student of the project group based on the individual performance in the viva-voce examination) as follows:

Phase I End Semester Examinations				
Report Evaluation		Viva-Voce		
Internal	External	Internal	External	*Guide
5	5	10	10	10

12.6 Special Supplementary Examination for project work (Phase I)

There will be a special supplementary viva-voce examination for the project work (Phase I), with a minimum duration of 3 weeks after the publication of respective end semester examination results. The pattern of evaluation will be the same as that of the end semester examination. Such appearance in special supplementary examination will be treated as another attempt and it will be reflected in the grade sheet.

12.6.1 A student will fail to secure a pass in project work (Phase I), if he/she

- I. Has not submitted project work (Phase I) report within stipulated date, before viva-voce.
- II. Has submitted the project work (Phase I) report within stipulated date, appeared for the viva-voce exam, but has failed to secure the passing marks.
- III. Has submitted the project work (Phase I) report within stipulated date before viva-voce but was absent for the viva-voce examination and has not obtained Withdrawal for the same (vide Clause 18).

12.6.2 A student who failed to secure a pass as per category 12.6.1- I, will be permitted to submit the project work (Phase I) report within a stipulated grace period of one month before the Special Supplementary project work (Phase I) Viva Voce examination and then he / she can attend the Special Supplementary project work (Phase I) Viva Voce examination provided he / she fulfilled the attendance requirements.

12.6.3 A student who failed to secure a pass as per category 12.6.1 - II, can attend a special supplementary Viva-Voce exam with improved project work (Phase I) report submitted as prescribed in the Special Supplementary examination schedule.

12.6.4 A student who failed to secure a pass as per category 12.6.1 - III, can attend a special supplementary Viva-Voce exam conducted in the presence of the External examiner with the project work (Phase I) report that was already submitted. There is no option to submit improved project work (Phase I) report. Special Supplementary Viva-Voce Exam is considered as reappearance with payment of exam fee.

12.6.5 Students who have withdrawn for valid reasons, as per Clause 18, are entitled to appear for the special supplementary examination and their appearance in the special supplementary examination will not be treated as another attempt. There is no option to submit improved project work (Phase I) report.

If a student eligible for improved project work (Phase I) report submission but does not make a submission within stipulated date, before viva-voce or if a student eligible for Special Supplementary Viva-voce Examination is absent for the same, or if a student fails to secure passing marks after the completion of Special Supplementary Viva-voce Examination, he/she shall repeat the course, only along with the regular students when the course is offered next. He / she has to fulfil attendance requirements, secure fresh internal assessment, submit the project work (Phase I) report and appear for the End Semester Viva-Voce examination.

12.7 Project Work Examination (Phase II)

The project work (Phase II) shall be evaluated for a maximum of 200 marks each. The Continuous Assessment will be done for 120 marks as per Clause 11.6.

The award of end semester marks for 80 shall be evaluated by both Internal Examiners and External Examiners, out of which the evaluation of the final report submitted by the student group shall carry a maximum of 20 marks (same marks must be awarded to every student of the project group) while the viva-voce examination shall carry 60 marks (awarded to each student of the project group based on the individual performance in the viva-voce examination) as follows:.

Phase II End Semester Examinations				
Report Evaluation		Viva-Voce		
Internal	External	Internal	External	*Guide
10	10	20	20	20

*Due to unavoidable reasons, in the absence of guide during the end semester examination, the Internal Examiner will award the marks for the concerned student.

12.8 Special Supplementary Examination for project work (Phase II)

There will be a special supplementary viva-voce examination for the project work (Phase II), with a minimum duration of 3 weeks after the publication of respective end semester examination results. The pattern of evaluation will be the same as that of the end semester examination. Such appearance in special supplementary examination will be treated as another attempt and it will be reflected in the grade sheet.

12.8.1 A student will fail to secure a pass in project work (Phase II), if he/she

- i. Has not submitted project work (Phase II) report within stipulated date, before viva-voce.
- ii. Has submitted the project work (Phase II) report within stipulated date, appeared for the viva-voce exam, but has failed to secure the passing marks.
- iii. Has submitted the project work (Phase II) report within stipulated date before viva-voce but was absent for the viva-voce examination and has not obtained Withdrawal for the same (vide Clause 18).

12.8.2 A student who failed to secure a pass as per category 12.8.1-i, will be permitted to submit the project work (Phase II) report within a stipulated grace period of one month before the Special Supplementary project work (Phase II) Viva Voce examination and then he / she can attend the Special Supplementary project work (Phase II) Viva Voce examination provided he / she fulfilled the attendance requirements.

12.8.3 A student who failed to secure a pass as per category 12.8.1 - ii, can attend a special supplementary Viva-Voce exam with improved project work (Phase II) report submitted as prescribed in the Special Supplementary examination schedule.

12.8.4 A student who failed to secure a pass as per category 12.8.1-iii, can attend a special supplementary Viva-Voce exam conducted in the presence of the External examiner with the project work (Phase II) report that was already submitted. There is no option to submit improved project work (Phase II) report. Special Supplementary Viva-Voce Exam is considered as reappearance with payment of exam fee.

12.8.5 Students who have withdrawn for valid reasons, as per Clause 18, are entitled to appear for the special supplementary examination and their appearance in the special supplementary examination will not be treated as another attempt. There is no option to submit improved project work (Phase II) report.

If a student eligible for improved project work (Phase II) report submission does not make a submission within stipulated date, before viva-voce or if a student eligible for Special Supplementary Viva-voce Examination is absent for the same, or if a student fails to secure passing marks after the completion of Special Supplementary Viva-voce Examination, he/she shall repeat the course, only along with the regular students when the course is offered next. He / she has to fulfil attendance requirements, secure fresh internal assessment, submit the project work (Phase II) report and appear for the End Semester Viva-Voce examination.

12.9 Experiential Project Based Learning Examination

This course is assessed through Continuous assessment mode as per vide clause 11.7.

If the concerned B.E. / B.Tech. CSE/IT/ECE student fails in the continuous assessment or takes a break and rejoins the programme at a later point of time in a semester other than the prescribed semesters identified for the course; he/she should opt for an open elective in lieu of this course.

12.10 Provision of a scribe:

The Institution will provide a scribe to students who are physically incapacitated to write the examination. Detailed guidelines will be available in the Office of the Controller of Examinations for perusal by students.

13. REQUIREMENTS FOR APPEARING FOR END SEMESTER EXAMINATION

A candidate shall normally be permitted to appear for End semester examination of the current semester if he / she has satisfied the semester completion requirements (vide clause 7) and has registered for examination in all courses of the current semester and all the courses, the student has not cleared in the

previous semesters by paying the prescribed fee, failing which, the candidates will not be permitted to attend the class for subsequent semester.

A candidate who has already appeared for any course in a semester and passed the examination is not entitled to reappear for the same course for improvement of letter grades / marks.

13.1 Special Supplementary Examination at the end of VI and VIII Semester

It is decided to conduct two Special Supplementary Examinations as follows:

- a. One Special Supplementary Examination shall be conducted immediately after the publication of the results of challenge valuation of the VI Semester Examination for candidates who have one or two arrears in theory / laboratory courses from I semester onwards to enable students to join the placement process. Students with three arrears and more are not eligible to appear for this special supplementary examination.
- b. Another Special Supplementary Examination may be conducted immediately after the publication of the results of challenge valuation of the VIII Semester for candidates who have one or two arrears in theory / laboratory courses only in the VII and VIII semesters. Students with three arrears and more in theory / laboratory courses are not eligible to appear for this special supplementary examination.
- c. The supplementary viva voce examination for project work (phase I and II) will be conducted as per clause 12.6 and 12.8 respectively.

Controller of Examination shall publish a schedule of special supplementary examination after the last date of registering for the examination. The pattern of evaluation will be the same as that of the end semester examination. Such appearance in special supplementary examination will be treated as another attempt and it will be reflected in the grade sheet. However, those students who have withdrawn any of the VIII semester courses including Project Work for valid reasons, as per Clause 18, are entitled to appear for the special supplementary examination and their appearance in the special supplementary examination will not be treated as another attempt.

14. PASSING REQUIREMENTS

14.1 Theory, Laboratory, Project Based Courses

A candidate who secures an aggregate mark of not less than 50% of total marks prescribed for the courses (Continuous Assessment + End semester examination) and also secures a minimum of 45% of the marks prescribed for the End semester examination in theory, laboratory courses (including Project work), shall be declared to have passed in the Examination. The passing requirement for Experiential Project Based Learning Course is 50% of the continuous assessment marks only.

14.2 Theory Courses with Laboratory Component (TCLC)

For a candidate undergoing a TCLC course consisting of two parts, say A and B, it is mandatory that the student must appear for both the parts (i.e., theory and laboratory) in the end semester examination.

- 14.2.1** A candidate who secures an aggregate mark of not less than 50% of the total marks prescribed for the courses (Continuous Assessment + End Semester Examination) and also secures a minimum of 45% of the marks prescribed for the End semester examination in both theory and laboratory components shall be declared to have passed in the examination.
- 14.2.2** If the candidate is absent or fails in both the parts of Theory Courses with Laboratory Component (TCLC), the candidate is declared as fail in both the parts and he / she should reappear (vide clause 4.2) for both, part A and B in the subsequent semesters.
- 14.2.3** If a candidate fails to secure a pass in a particular part of TCLC course, it is mandatory that he / she shall reappear vide Clause 4.2 for that part alone during the next semester when examination is conducted for that part of the course. It is mandatory that he / she should continue to reappear for the examination till he / she secures a pass mark.

14.3 Consideration of Internal marks

If a candidate fails to secure a pass in a particular course, it is mandatory that he / she shall register and reappear for the examination in that course during the next semester when examination is conducted in that course. It is mandatory that he / she should continue to register and reappear for the examination till he / she secures a pass mark.

14.3.1 The internal assessment marks obtained by the candidate in the first appearance shall be retained and considered valid for all subsequent attempts till the candidate secures a pass as per clause 14.1 and 14.2. However, from the third appearance onwards if a candidate fails to obtain pass marks (Internal Assessment + End Semester Examination) as per clause 14.1 and 14.2, then the passing requirement shall be as per clause 14.3.2.

14.3.2 For arrear students, from their third appearance onwards, they should secure a minimum of 50% marks prescribed for the course in the END Semester Examination alone.

14.4 Evaluation of Answer Scripts

The system of evaluation of answer scripts for theory courses is “Double Valuation” i.e., valuation by two examiners independently. Since the answer scripts are subject to double valuation, there is no provision for revaluation. Even after a double valuation, candidates are given the right to challenge the valuation. Challenging the valuation is permitted for those students who have applied for obtaining photocopy of answer script. Students can apply for challenge valuation in a theory course, within 2 weeks from the declaration of results, on payment of a prescribed fee along with prescribed application form submitted to the Controller of Examinations (COE) through the Heads of Departments.

Copies of answer scripts for theory courses can be obtained from the Office of the Controller of Examinations on payment of a prescribed fee specified for this purpose through proper application. The copy of answer script is to be valued by a competent authority and the valued script should be submitted to Controller of Examination’s (COE’s) office along with the prescribed fee for challenging the valuation. The COE will arrange for the challenge valuation and the publications of the results.

Challenging is not permitted for Laboratory Courses, Mini projects, Seminar, Project work, laboratory components of TCLC courses and Experimental based learning.

14.4.1 After double valuation of each answer script, the average of two Valuation marks (ie, V_1 – 1st valuation mark; V_2 – 2nd valuation mark of the same QP code) is taken as the final mark of the Candidate in that paper.

14.4.2 If the variation in grand total marks obtained in two valuations is 15 and above, then the answer script (of the same QP code) shall undergo a third valuation (ie, V_3 – 3rd valuation). In the case of answer scripts going for third valuation, the average of highest two marks shall be taken as the final mark.

14.4.3 If the answer script has undergone for Challenging Valuation, two experts will jointly value the answer script by thoroughly analyzing answers written by the candidate and provide the Challenging Valuation Mark (CVM). The CVM will be compared with the previous two or three valuation marks. The final mark is computed as per any one of the following criteria whichever is applicable:

Criterion 1: Average of HIGHEST TWO among V_1 , V_2 and CVM marks

OR

Criterion 2: Average of HIGHEST TWO among V_1 , V_2 , V_3 and CVM marks

14.3.3.1 If the obtained grade in Challenging Valuation (CVM) is less than the already published grade, then there will be No Change in Grade.

Note: Wherever this Academic Regulation is silent on any particular aspect of Examination, the provisions made in the Examination manual of MEPCO shall apply mutatis mutandis.

15. AWARD OF LETTER GRADES

If students strength in a theory course ≥ 30 , then the award of letter grades will be decided based on relative grading principle. The relative grading is applicable to ONLY those students who have passed the examination as per the minimum passing requirements. For those students who have not passed the examination, Reappearance (U) shall be awarded.

For those students who have passed the course, the relative grading shall be done using the software provided by Anna University, Chennai.

However, for theory courses where the students' strength is less than 30 and for all laboratory courses, mini project, project work and experimental based learning examinations, the fixed grading shall be followed with the grade range as specified below:

Letter Grade	O	A+	A	B+	B	C	U
% Marks	91 - 100	81 - 90	71 - 80	61 - 70	56 - 60	50 - 55	< 50

The performance of a student shall be reported using letter grades, each carrying certain points as detailed below:

Letter Grade	Grade Points
O (Outstanding)	10
A + (Excellent)	9
A (Very Good)	8
B + (Good)	7
B (Average)	6
C (Satisfactory)	5
U (Re-appearance)	0
SA (Shortage of Attendance)	0
W (Withdrawal)	0

A student is deemed to have passed and acquired the corresponding credits in a particular course if he/she obtains any one of the following grades: "O", "A+", "A", "B+", "B", "C". 'SA' denotes shortage of attendance for the student and hence prevented from writing the end semester examinations. 'SA' will appear only in the result sheet.

"U" denotes that the student has failed to pass in that course. "W" denotes withdrawal from the exam for the particular course. The grades "U" and "W" will figure both in the Grade Sheet as well as in the Result Sheet. In both cases, the student has to appear for the End Semester Examinations.

If the grade "U" is given to Theory Courses / Laboratory Courses / EEC (Employability Enhancement Courses), it is not required to satisfy the attendance requirements but has to appear for the end semester examination and fulfill the passing requirements to earn a pass in the respective courses.

16. SGPA AND CGPA CALCULATION

After the results are declared, Grade Sheets will be issued to each student which will contain the following details:

- The list of courses enrolled during the semester and the grades scored.
- The Semester Grade Point Average (SGPA) and
- The Cumulative Grade Point Average (CGPA) of all courses enrolled from first semester (third semester in case of LES) onwards.

During each semester, the list of courses enrolled and the grades scored in each course are used to compute the Semester Grade Point Average (SGPA). SGPA is the ratio of the sum of the products of the number of credits of courses registered and the grade points corresponding to the grades scored in those courses, taken for all the courses, to the sum of the number of credits of all the courses in the semester.

$$SGPA = \frac{\sum_{i=1}^n C_i GP_i}{\sum_{i=1}^n C_i}$$

- C - Credits assigned to each course
- GP - grade point corresponding to the letter grade obtained for each course
- n - Number of Courses successfully cleared during the particular semester in the case of SGPA and during all the semesters in the case of CGPA

CGPA will be calculated in a similar manner, considering all the courses enrolled from first semester (third semester in case of LES). "U", "SA", "W" and "AU" grades will be excluded for calculating SGPA and CGPA.

16.1 Conversion of CGPA to percentage of marks

The equivalent percentage of marks shall be calculated from the Cumulative Grade Point Average (CGPA) by multiplying with a factor of 10.

17. ELIGIBILITY FOR THE AWARD OF DEGREE

17.1. A student shall be declared to be eligible for the award of the B.E. / B.Tech. Degree provided,

- i. The student has successfully gained the required number of total credits as specified in the curriculum corresponding to his/her programme within the stipulated time.
- ii. The student has successfully completed the programme requirements and has passed all the courses prescribed in all the 8 semesters (six semesters in case of LES) within a maximum period of 7 years (six years in case of LES) reckoned from the commencement of the first semester (third semester in case of LES) to which the candidate was admitted.
- iii. The student has successfully completed any additional courses prescribed by the Chairperson, Academic Council whenever, any candidate is readmitted with regulations other than MEPCO R-2023 (vide Clause 19.2).
- iv. The student has no disciplinary action pending against him/her.
- v. The award of Degree shall be approved by the Syndicate of the University.

17.2 Classification of the Degree Awarded

17.2.1 First Class with Distinction :

A candidate who satisfies the following conditions shall be declared to have passed the examination in First class with Distinction for the programs B.E / B.Tech Regular, B.E / B.Tech with Specialisation, B.E / B.Tech Honours and B.E / B.Tech Minor with specialisation:

- Should have passed the End semester examinations in all the courses of all the eight semesters (six semesters in case of LES) in his/her First Appearance within four years (three years in case of LES).
- Should have secured a CGPA of not less than 8.5
- Should NOT have been prevented from writing end semester examination due to lack of attendance in any of the semesters.
- Withdrawal from examination (vide Clause 18) will not be construed as an appearance.
- One year authorized break of study (if availed of) is permitted in addition to four years (three years in case of LES) for award of First class with Distinction.

17.2.2 First Class Honours with Specialization

A candidate who satisfies the following conditions shall be declared to have passed the examination in First Class Honours with Specialization.

- Should have passed the End semester examinations in any 6 courses belonging to a particular emerging area which is eligible as Honours for his / her Programme (vide Clause 5) in first appearance.
- Should have secured a CGPA of not less than 7.5
- The student has to clear all the courses (Both Regular and Honours) in first appearance.
- Withdrawal from examination (vide Clause 18) will not be construed as an appearance.
- One year authorized break of study (if availed of) is permitted in addition to four years (three years in case of LES) for award of First Class Honours with Specialization.

17.2.3 First Class Honours

A candidate who satisfies the following conditions shall be declared to have passed the examination in First Class Honours.

- Should have passed the End semester examinations in any 6 courses across any of the emerging areas which are eligible as Honours for his / her programme (vide Clause 5) in first appearance.
- Should have secured a CGPA of not less than 7.5
- The student has to clear all the courses (Both Regular and Honours) in first appearance.
- Withdrawal from examination (vide Clause 18) will not be construed as an appearance.
- One year authorized break of study (if availed of) is permitted in addition to four years (three years in case of LES) for award of First Class Honours.

17.2.4 First Class Minor with Specialization

A candidate who satisfies the following conditions shall be declared to have passed the examination in First Class Minor with Specialization.

- Should have passed the End semester examinations in any 6 courses belonging to a particular emerging area which is eligible as Minor for his / her programme (vide Clause 5).
- Should have passed the examination in all the courses of all eight semesters (6 semesters in the case of LES) within five years (four years in the case of LES).
- Should have secured a CGPA of not less than 6.5
- Withdrawal from examination (vide Clause 18) will not be construed as an appearance.
- One year authorized break of study (if availed of) or prevention from writing the End Semester examination due to lack of attendance (if applicable) is included in the duration of five years (four years in the case of LES) for the award of First class minor with specialization.

17.2.5 First Class:

A student who satisfies the following conditions shall be declared to have passed the examination in First class:

- Should have passed the examination in all the courses of all eight semesters (6 semesters in the case of LES) within five years (four years in the case of LES).
- Should have secured a CGPA of not less than 6.5
- Withdrawal from examination (vide Clause 18) will not be construed as an appearance.
- One year authorized break of study (if availed of) or prevention from writing the End Semester examination due to lack of attendance (if applicable) is included in the duration of five years (four years in the case of LES) for the award of First class.

17.2.6 Second Class:

All other candidates (not covered in Clause 17.2.1 to Clause 17.2.5) who qualify for the award of the degree (vide Clause 17.1) shall be declared to have passed the examination in Second Class.

17.2.6.1 A student earned additional 18 credits as per clause 5.1 and 5.2, but does not satisfy the conditions mentioned in 17.2.1 to 17.2.3, shall not be awarded B.E./B.Tech. Honours. In such cases, the grade statement show the additional courses studied and those courses shall not be considered for CGPA computation. In such cases if the student becomes eligible for the first class as per 17.2.5, while computing CGPA without taking into account the additional courses studied, the student shall be awarded B.E. / B.Tech. in First class as per 17.2.5 only.

17.2.7 A candidate who is absent for semester examination in a course / project work after having registered for the same shall be considered to have appeared in that examination for the purpose of classification.

17.3 Ranks Awarded to Students

Ranks are awarded for each programme based on the following criteria.

- The number of ranks awarded for each programme is 10% of the students' strength of that programme.
- The rank of a student is determined from CGPA. However, students who have scored less than 7.5 CGPA will not be eligible for a rank.
- Students should have passed the End semester examination in all the courses of all the eight semesters (six semesters in case of LES) in his/her First Appearance without any break of study.
- However, for other students, a statement indicating their position in the programme will be given on request, based on the CGPA, provided they have passed all the courses within four years (three years in case of LES) of their study.

18. PROVISION FOR WITHDRAWAL FROM EXAMINATION

18.1 A candidate may, for valid reasons, (medically unfit/unexpected family situations) be granted permission to withdraw from appearing for the examination in any course(s) or all courses in any one of the semester examination during the entire duration of the degree programme.

However, if a candidate requests to withdraw from appearing for the examination in any course or courses for second time during his / her course of study due to valid reasons, he / she shall apply to the Principal through the Head of the Department.

- 18.2** Such withdrawal shall be permitted only once during the entire period of study of the degree programme.
- 18.3** Withdrawal application shall be valid only if the candidate is otherwise eligible to write the examination (Clause 6) and if it is made within TEN days before the commencement of the examination in that course or courses and also recommended by the Head of the Department and approved by the Head of the Institution with intimation to COE.
- 18.3.1** Notwithstanding the requirement of mandatory TEN days' notice, applications for withdrawal for special cases under extraordinary conditions will be considered based on the merit of the case.
- 18.4** Withdrawal shall not be construed as an appearance for deciding the eligibility of a candidate for First Class with Distinction and First Class.
- 18.5** Withdrawal is NOT permitted for arrears examinations of the previous semesters.
- 18.6** Candidates shall appear, for all the courses withdrawn (vide Clause 18.1), during the examination conducted in the subsequent semester itself.

19. AUTHORIZED BREAK OF STUDY FROM A PROGRAMME

- 19.1** A candidate is normally not permitted to temporarily break the period of study. However, if a candidate happens to discontinue the programme temporarily in the middle of duration of study for valid reasons (such as accident or hospitalization due to prolonged ill health) and to rejoin the programme in a later semester, he / she shall apply to the Chairperson, Academic Council in advance, in any case, not later than the last date for registering for the semester in question, through the Head of the Department stating the reasons.
- 19.2** The candidate permitted to rejoin the programme after the break shall be governed by the Curricula and regulations in force at the time of rejoining.
- 19.3** The total period for completion of the programme reckoned from, the commencement of the first semester (third semester in case of LES) to which the candidate was admitted shall not exceed the maximum period specified in clause 6.1 irrespective of the period of break of study in order that he / she may be eligible for the award of the degree (vide Clause 17).
- 19.4** If any student is detained for want of requisite attendance, progress and good conduct, the period spent in that semester shall not be considered as permitted 'Break of Study' and Clause 19.3 is not applicable for this case.
- 19.5** In case of any valid reasons for extension of Break of Study, it may be granted by the approval of the Chairperson, Academic Council for a period not more than 1 year in addition to the earlier authorized Break of Study. Such extended break of study shall be counted for the purpose of classification of degree (vide Clause 17.2).
- 19.6** If the candidate has not reported back to the department, even after the extended Break of Study, the name of the candidate shall be deleted permanently from the college enrollment. Such candidates are not entitled to seek readmission under any circumstances.

20. INDUSTRIAL VISIT / INDUSTRIAL TRAINING

Every student is expected to undergo Industrial visits every year for a minimum period of one / two / three / four days respectively based on the year of study. Heads of Departments shall take efforts to arrange at least one industrial visit in a year as above. The students may undergo industrial training for a total period of 2–4 weeks during summer / winter vacation.

21. PERSONALITY AND CHARACTER DEVELOPMENT

All students shall enroll, on admission, in the personality and character development programmes like NCC, NSS, NSO, YRC, Yoga and undergo training for about 80 hours inclusive of attending camps of about

seven days. The training shall include classes on hygiene and health awareness and also training in first-aid.

- National Cadet Corps (NCC) will have about 20 parades per year.
- National Service Scheme (NSS) will have social service activities in and around the College / Institution.
- National Sports Organization (NSO) will have sports, Games, Drills and Physical exercises.
- Youth Red Cross (YRC) will have activities related to social services in and around college/institutions. While the training activities will normally be during weekends, the camp will normally be during vacation period.
- Yoga for Youth Empowerment program will be conducted for one year during I / II year of study to maintain harmony of body and spirit. Yoga sessions will be conducted to increase the attention span, to improve memory and to sharpen the focus of the students. One meditation session for 5 minutes every morning will also be conducted before starting of regular classes.

Every student shall put in a minimum of 75% attendance in the training and attend the camp compulsorily. The training and camp shall be completed during the first year (second year in case of LES) of the programme. However, for valid reasons, the Head of the Institution may permit a student to complete this requirement in the second year (third year in case of LES) also.

22. DISCIPLINE

Every student is required to observe disciplined and decorous behavior both inside and outside the Institution and not to indulge in any activity which will tend to tarnish the image of the Institution.

In the event of an act of indiscipline being reported, the Principal shall constitute a disciplinary committee to enquire the misbehavior.

If a student indulges in malpractice in any of the End semester examination / Continuous Assessment, he / she shall be liable for punitive action as prescribed by the Institution from time to time.

Ragging is not at all allowed. Punitive actions will be taken against the students involved in ragging as per the government norms.

23. AUDIT COURSE

UG students can audit courses from UG / PG curriculum based on their areas of interest and is permitted under the following conditions:

- Students with CGPA 7.5 and above will be permitted to 'Audit' the course. However, this would be restricted to a maximum of 2 courses during the entire period of the programme.
- The students have to register for the courses to be audited through Course Registration Form while registering for the semester. The course will not carry any credits. The word "Audit" would be specially mentioned in the remarks column of the student's course registration form. Prior permission of the Course Instructor and the Heads of the Respective Departments required.
- The students would be required to have minimum 75% attendance in the Audit course.
- The 'AU' grade would be awarded, if the attendance is satisfactory and requirements set out by the course instructor are met. Students will be expected to complete the in-semester assessments. If the attendance and performance is not satisfactory, the course will not appear at all in the Grade Sheet.
- Students can audit a course provided the course is offered and the time table slot permits.
- The course done by auditing will not be considered for the purposes of calculation of SGPA / CGPA but will be reflected in the Semester Grade Sheet as Audit Course.

24. VALUE ADDED COURSES

In order to prepare the students to meet the challenges of the global work environment, Value Added Courses are offered to bridge the gap between the curriculum and the requirements of the industry. Value Added Courses which may lead to international certification are designed and offered by each department for the benefit of the students. Each Value Added Course is assigned 1 Credit. The credits earned through Value Added Courses will be considered for CGPA calculation. The performance of the students in the Value Added Courses is evaluated through examination for 100 marks. If a student gets a pass mark in the Value Added Course examination, that VAC will be entered in respective semester grade sheet and the credits earned will be eligible for credit calculations and included in CGPA.

25. INTERNATIONAL CERTIFICATION

International certification establishes an industry standard by which students are evaluated through a fair, comprehensive test of their knowledge in the areas related to their disciplines. Being certified makes the student, a highly competent professional. After attending the VAC as per Clause 24, students can appear for the respective international certification examinations. For example, certificate courses like Certified LabVIEW Associate Developer (CLAD), Certified LabVIEW Developer (CLD), SIEMENS PLC Programming, Cisco Certified Network Associate (CCNA), RedHat certification, Cloud Computing certification, Oracle certification, Microsoft certification, ANSYS certification, Certified SOLIDWORKS Associate (CSWA), Building Information Modeling (BIM) and other relevant courses as suggested and approved by HoD and BoS may be offered. The students should mandatorily attend one international certification course during their course of study. Students who have successfully received International Certification will earn 1 credit. The credits earned will be considered for CGPA calculation.

If a student gets a pass mark in the International certification, then the credits earned will be mentioned in the grade sheet or else, it will not be reflected in the grade sheet.

26. ONLINE COURSES

Online Courses (OLC) are courses that would be offered by premier institutions like IITs, IISc through online platform (NPTEL / GIAN / MOOCS / COURSERA, etc.) and suggested by the concerned Department. The list of online courses will be decided by the Department beyond the prescribed curriculum and approved by the respective Board of Studies (BoS). The successful completion of each online course of at least eight week duration shall be given 1 credit, while courses of duration twelve week and above shall be given 1.5 credits. The students should attend minimum of one and maximum of two online courses per semester from third to sixth semester of study. It is optional during other semesters. Details of online courses successfully completed by the students (from I to VIII semesters) shall be maintained in a separate Additional Credit Bank. If a course from the bank is used for dropping (Clause 31), only such courses alone will be considered for SGPA and CGPA calculation. The same shall be updated IN THE CONCERNED Semester (VI/VII/VIII) grade statement. The unused courses and their credits available in the Additional Credit Bank will not be considered for CGPA calculation, however they will be reflected in the Consolidated Grade Statement.

27. SPECIAL COURSES

Interested Students can opt for special courses of 15 or 30 hours duration, approved by the academic council. Special courses will be conducted by experts from industry/research organizations / higher learning institutions. Each 15 Hour special course is assigned 1 credit, while each 30 Hour Special Course is assigned 2 credits. The credits earned through special course will not be considered for CGPA calculation. Students can complete such special course during the semesters 3 to 8 as and when these courses are conducted by different departments. A student is also permitted to register for special course of other departments, provided the student has fulfilled the necessary pre-requisites of the course being offered and subject to the approval of both the heads of departments. A student can register for a maximum of one special course per semester. The special course will be evaluated by the respective course coordinator within 10 days of the completion of the programme.

If a student gets a pass mark in the special course, then the credits earned will be mentioned in the grade sheet or else, it will not be reflected in the grade sheet.

28. PRO-INTERNSHIP

Pro-internship (Fast Track) scheme is designed specifically for students with good academic credentials to take up internships. It presents a unique opportunity for the students to complete the theory courses of the program in seven semesters. Pro-internship allows students to take up the VIII semester theory courses in VI and VII semesters itself. Students can go for internship during the VIII semester and complete the VIII semester Project Work in well reputed Industries / organizations offering Placement to the students / R & D organizations / Premier Institutions both in India and Abroad, as part of their internship. This is possible only if the Industries / organizations offering Placement to the students / R & D organizations / Premier Institutions allow Internship program with Project and the internship coordinators of the Department concerned are notified well in advance.

28.1 Eligibility

The guidelines presented below will be the eligibility for the students to opt for Fast track

- 1) Students who are in the VI semester with a minimum CGPA of 7.5 up to V semester.
- 2) Those who have completed all the courses up to V semester.
- 3) However only students who complete all the courses up to VII semester will be allowed to proceed with the Internship cum Project.

28.2 Application to the Pro – Internship (Fast Track) Scheme:

Students eligible for Fast Track scheme should apply for the scheme to their HoD within 7 days from the date of publication of V semester results in the VI semester of their study.

28.3 Guidelines to be followed:

- 1) Fast Track students should take at least one VIII semester theory course in VI semester. The rest of the VIII semester theory courses can be completed in VII semester.
- 2) The Fast Track option requires students to attend classes of 4 periods duration per week for the VIII semester courses during the VI and VII semesters of their study. If required, students need to attend additional classes during holidays to meet the curriculum requirement.
- 3) Students will be permitted to start the internship only after the seventh semester end terminal examinations
- 4) Students doing the internship should attend the Project Reviews in the scheduled dates. After completion of the internship cum project, the final summative evaluation will be conducted with a view to assign the final score.

28.4 Terms and conditions for internship:

1. The internship should be carried out in a single organization.

2. It should be completed within one semester.
3. All arrangements should be specified in the internship agreement.
4. During long leave period and closure of Industries/R&D organization/Premier Institutions for any reason, during the internship period, the students concerned should continue the work only in the college.
5. Students should adhere to the rules and regulations of the college and Industries/R&D organization/Premier Institutions during the Internship cum Project period.

28.5 Termination of the Internship

1. In the event of prolonged absence of student during the internship or
2. If the student has to discontinue the internship due to unforeseen circumstances during that period or
3. If the performance of the work done by the student in the Industries/R&D organization/Premier Institutions is not satisfactory during the Project Review or
4. If the student violates the college rules and regulations during the Internship cum Project period,

Then the Internship cum Project will be terminated for these students, after due intimation to the company. In such a situation, the VIII Semester Project will be in jeopardy, i.e. these students have to repeat their VIII Semester project in the next academic year.

29. Research Paper Publications

The UG students are encouraged to publish their project and research works in reputed journals. An article published in SCI journal shall be given 2 credits, whereas an article published in SCOPUS journal shall be given 1 credit. The grades will be awarded as given below.

Publications in	Letter Grade
SCI (WoS) – Q1 Ranked Journals	O
SCI (WoS) – Q2 Ranked Journals	A+
SCI (WoS) – Q3 Ranked Journals	A
SCI (WoS) – Q4 Ranked Journals	B+
SCOPUS – Q1 Ranked Journals	O
SCOPUS – Q2 Ranked Journals	A+
SCOPUS – Q3 Ranked Journals	A
SCOPUS – Q4 Ranked Journals	B+

If a student publishes an international journal paper as above with a valid DOI number / volume number, page number before two weeks from publication of results of any end semester examination, the hard copy of the research paper publication should be submitted. Then only the credits earned will be mentioned in the grade sheet or else, it will not be reflected in the grade sheet. The credits earned will not be considered for CGPA calculation.

30. INTERNSHIP/FELLOWSHIP

Internship / Fellowship is designed in order to provide industry / R&D exposure to all the students. Students can opt for internship / Fellowship during their III to VI semester study in well reputed Industries / organizations offering Placement to the students / R & D organizations / Premier Institutions both in India and Abroad. This is possible only if the students avail prior permission from the Head of the Department as well the industry counterpart. The terms and conditions and termination of the internship specified in Clause 28.4 and 28.5 are applicable here also.

Guidelines for Internship:

1. Students need to apply for internship through the Head of the Department.
2. Only reputed industries / Organizations are to be considered by the students while applying for internship.

3. The Credits earned by the students for the internship programme is as follows:
 For 4 week continuous internship in a single organization – 1 Credit
 For 8 week continuous internship in a single organization – 3 Credits
4. The students are not allowed to split up the internship periods in multiple companies.
5. The credits earned in this scheme will be included in the CGPA calculation.
6. Also, the credits earned in this scheme would be used for dropping any one Allied/Open elective.
7. In case, if a student is not able to continue the internship due to any inevitable reason, this scheme is treated as cancelled for him/her.
8. The progress of the internship carried out by the student needs to be monitored by the concerned Supervisor by visiting the industry at least once.
9. A committee constituted by the HoD will evaluate the work done by the students through their report and presentation. The grades will be awarded as per Clause 15.

31. DROPPING OF CE, OE COURSES

The students have the option of dropping a maximum of 2 elective courses of eighth semester, if the student has earned enough credits through the International Certification (vide Clause 4.5 (iv)) with not less than B+ Grade, Value Added Courses (vide Clause 24) with not less than B+ Grade, Online Courses (vide Clause 26) with not less than B+ Grade and Internship (vide Clause 30).

The students should submit the requisition to drop the course in prescribed form for approval by the Head of Department before start of the semester.

While dropping the elective courses, the following points need to be considered.

- 31.1** Students who have successfully completed online courses in other disciplines as approved by BoS specified for the same semester and earned credits, can drop Open Elective courses.
- 31.2** Students who have successfully completed online courses in their core disciplines as approved by BoS specified for the same semester and earned credits can drop Core / Open Elective courses.
- 31.3** Students who have successfully completed online courses approved by BoS for immediate previous semester and earn credits, can also drop courses as per clauses 31.1 and 31.2.
- 31.4** Students who have successfully completed online courses approved by BoS for lower semesters and earn credits, cannot drop courses as per clauses 31.1 and 31.2.

32. REVISION OF REGULATIONS AND CURRICULUM

The Institution may from time to time revise, amend or change the Regulations, scheme of examinations and syllabi if found necessary. However, a Standing Committee shall be formed by Principal on need basis for any special case, which falls beyond the above rules and regulations. They will offer suitable interpretations/clarifications/amendments required for special case. All such exceptions will have to be ratified in the next meeting of the Academic Council. The decision of the Academic Council is final.

ANNEXURE

A candidate may be offered one of the following programmes of study approved by the University. A candidate of a programme should earn the minimum credits specified against it to enable him/her to be eligible to be awarded the degree. The total credit for the following programmes varies between a minimum of 160 to a maximum of 180.

Name of the Programme		Minimum Credits
		Regular
1. FACULTY OF CIVIL ENGINEERING		
i)	B. E. Civil Engineering	171
2. FACULTY OF ELECTRICAL AND ELECTRONICS ENGINEERING		
i)	B. E. Electrical and Electronics Engineering	171
3. FACULTY OF ELECTRONICS AND COMMUNICATION ENGINEERING		
i)	B. E. Electronics and Communication Engineering	171
4. FACULTY OF COMPUTER SCIENCE AND ENGINEERING		
i)	B. E. Computer Science and Engineering	171
5. FACULTY OF MECHANICAL ENGINEERING		
i)	B. E. Mechanical Engineering	170
6. FACULTY OF INFORMATION TECHNOLOGY		
i)	B. Tech. Information Technology	171
7. FACULTY OF BIOTECHNOLOGY		
i)	B.Tech. Biotechnology	171
8. FACULTY OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE		
i)	B.Tech Artificial Intelligence and Data Science	171
9. FACULTY OF BIO MEDICAL ENGINEERING		
i)	B.E Bio Medical Engineering	171